

MICROMOLD · DESIGN DECISIONS REGISTER

MicroMold design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/micromold		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

MicroMold design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	The design changes made for construction	Accept, or change any of P1 to P15	Accept	Every component; the build plan is drawn to them	MMD-DDR-003, Table 1
2	How flake is loaded: with the plunger raised its tip is only 8 mm above the funnel rim	(a) pull the ball-lock pin and lift the plunger out to load, refit it to tamp and inject; (b) a press with 40 mm more ram travel and a taller column (handle top about 1,134 mm, over R9); (c) a side loading chute	(a): no new part; the plunger's top end is about 94 °C, handled in the gloves already required	The operating procedure; nothing in the build changes for (a)	MMD-DDR-003, A1
3	The lift screw needs a 25 mm hole through the bench and drops up to 50 mm below it	(a) accept; (b) a 60 mm riser frame under the base plate (handle top over 1.1 m)	(a)	Step 13 (bench hole)	MMD-DDR-003, A2
4	Handle stop or pull limit against overload (a 700 N pull gives about 35 MPa, 1.35 times the press rating)	(a) a stop on the ratchet arc; (b) a torque-limiting socket on the ratchet adapter; (c) written procedure and the shield only	(b): it limits force whoever pulls	Ratchet adapter (section 3.6)	MMD-DDR-003, A3; docs/REVIEW.md , TRL 3 safety concerns
5	The specific first co-design partner and its city or region	Any group that already shreds HDPE or PP (D10)	None yet	Not part of the TRL 3 build; needed at TRL 4	MMD-DDR-001, O1
6	First product molds after the test plaque	To come from co-design	None yet	A new mold sketch per product; the press is unchanged	MMD-DDR-001, O2
7	Appearance model differences from the engineering model (2026-09-26): round duct stub, guide rods under the table, perforated shield window, full mold bolts, control box front, illustrative display values	Accept each, or change the appearance model	Accept 1, 3, 5 and 6 (the build plan also uses a round stub); for 2 and 4 the constructable design differs (no guide rods; cap screws, no nuts), so update the appearance model to it	The renders only, not the build	docs/REVIEW.md , 2026-09-26, items 1 to 6

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The arbor press: a 28 mm ram 255 mm or longer with 150 mm of engaged travel; a head that can be cut from its frame with a flat back face and enough metal for four M10 holes 20 deep; the head's mass (8 kg assumed)	Sets the stroke, the head fixing and R9's mass margin (1.0 kg)	MMD-DDR-003, P1; MMD-CAL-001 [K1]
2	The press's lever hub: its outside diameter and bore, to take the welded square drive	The ratchet adapter is made from it	MMD-DDR-003, P13
3	The load cell is about 56 mm across, has a top M12 stud and a base flange with three M5 through holes (46 mm circle assumed)	The spacer's tapped holes are drilled to match it	MMD-DDR-003, P6
4	The handwheel's hub takes a Tr20 x 4 nut (about 36 mm across)	The handwheel nut is made from it	MMD-DDR-003, P7
5	The mold cooling fan's corner holes are on a 105 mm square	The fan bracket's holes	MMD-DDR-003, P10
6	The nozzle heater band fits a 24 mm body and is no more than 18 mm wide	The nozzle body length	MMD-DDR-003, P4
7	The bench top's thickness and the clear space under it (60 mm or more below the hole)	The lift screw drops through it	MMD-DDR-003, P7

■ Value engineering

Value-engineering target: USD 520 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 544 for the press (USD 24 over the target); USD 638 with the test mold, which is tooling outside the target (USD 94, R8). Main cost drivers and savings worth trying:

- The largest lines are the arbor press head with column tube and ratchet (USD 110), the control box (USD 55), the fume hood and duct fan (USD 48), the barrel (USD 45), the load cell and display (USD 35) and the lift table, screw and handwheel (USD 30).
- Making the design constructable added USD 37: the construction parts of line 19 (USD 30: head mounting plate and screws, plunger coupling and ball-lock pin, ratchet adapter, hood arm and rivnuts), the bracket's back plate and cap screws (USD 4) and the shield's hinges and latch (USD 3). The mold rose USD 4 for screws, inserts and dowels.
- Savings worth trying: a used arbor press (often under half the new price); counting the duct fan and duct (USD 48) as workshop extraction shared with other machines, as the molds are counted as tooling, which would bring the press to USD 496; buying the two PID controllers and relays as a pre-wired dual-zone controller; and a ball-lock pin replaced by a plain clevis pin and R-clip (about USD 4 less).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: rack-and-pinion drive from a 1 t arbor press on a taller column, vertical barrel with the mold below, plunger injection, two PID zones, bolted two-plate molds on a screw lift table, 10 kN load cell, test plaque first, molds as tooling, side hood with duct fan, a partner that already shreds HDPE or PP	Amish: "i accept all your recommendations, go with them across all repos."	MMD-DDR-001, MMD-DDR-002
2026-09-25	N1 to N5: budget USD 500 for the press, 40 kg mass target, two 300 W barrel bands, a mold cooling fan, injection-grade flake only	Amish, same instruction	MMD-DDR-002
2026-09-26	Budget USD 520 to cover the priced BOM (O3, option (a))	Amish: "i approve all the budget items."	MMD-DDR-002 v0.2
2026-09-30	Make the design physically buildable as the build plan is drawn; changes recorded as draft for review	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	MMD-DDR-003 (its changes are open, item 1 above)
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	This register
2026-10-01	<code>budget_usd</code> is a hypothetical value-engineering target, not a spending limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register; MMD-REQ-001 R14